1a. Conductividad del alambre del cobre.

V = 10,00 v L = 100,00 m

A = 2,151 mm^2 P = 111,44 w

P = I V V = I R $R = \rho \frac{L}{A}$

$\Rightarrow$ $\boxed{\sigma = \frac{PL}{AV^2}}$ $\Rightarrow$ $\boxed{\sigma = 5{,}181x10^7\ S\ m^{-1}}$

1b. Tiempo promedio para recorrer un cm.

L = v t J = n e v $n = \frac{\#\ de\ partículas}{Volumen}$

$\Rightarrow$ $\boxed{t = \frac{n\, e\, L^2}{\sigma\, V}}$ $\Rightarrow$ $\boxed{t = 26{,}20\ s}$

2a. Resistencia entre las bases.

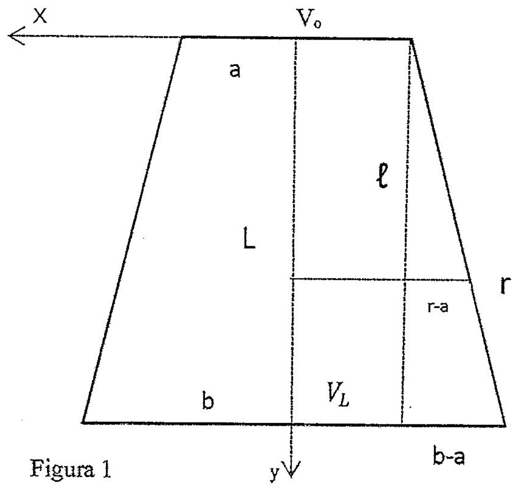


Figura 1

De la figura 1: $\boxed{\frac{r-a}{b-a} = \frac{\ell}{L}}$

$\Rightarrow$ $\boxed{r = a + \left(\frac{b-a}{L}\right)\ell}$

Además: $J = \frac{I}{\pi r^2} = \sigma E \qquad E = -\frac{dV}{dy}$

E dy = - dV $\qquad E = \frac{I}{\sigma \pi r^2}$

$$\frac{I\, dy}{\sigma \pi r^2} = -dV$$

$$\frac{I}{\sigma \pi} \frac{d\ell}{\left[a + \left(\frac{b-a}{L}\right)\ell\right]^2} = -dV$$

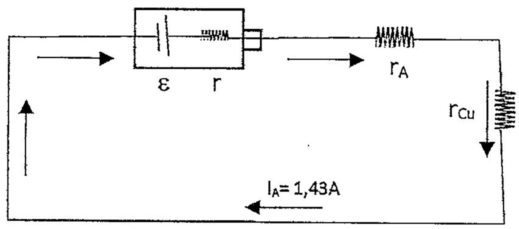
$$\boxed{\frac{I}{\sigma \pi} \int_0^L \frac{d\ell}{\left[a + \left(\frac{b-a}{L}\right)\ell\right]^2} = -\int_{V_0}^{V_L} dV}$$

$\Rightarrow$ $\boxed{R = \rho \frac{L}{\pi ab}}$

2b. **Valor de la resistencia cuando a=b.**

$$\boxed{R = \rho \frac{L}{\pi r^2}}$$

3. fem y resistencia de la pila.



ε - 1,43 r – 1,43 r_A – 1,43 r_{Cu} = 0

Ídem para la resistencia de aluminio,

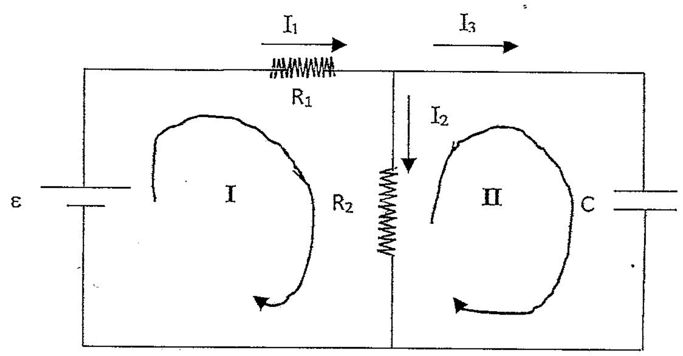
ε - r – r_A – r_{Al} = 0

r_A = 0,05 Ω $\qquad r_{Cu} = \rho_{Cu} \frac{L_{Cu}}{A_{Cu}} \qquad r_{Al} = \rho_{Al} \frac{L_{Al}}{A_{Al}}$

r_{Cu} = 0,850 Ω $\qquad r_{Al}$ = 1,45 Ω

Entonces: $\boxed{r = 0{,}50\ \Omega}$ $\boxed{\varepsilon = 2{,}00\ V}$

4. Análisis del circuito RC



ε=10 V R1=2 KΩ R2=3 MΩ C= 1 μF

$$Malla\ I:\ \varepsilon = I_1R_1 + I_2R_2 \quad Malla\ II:\ \frac{q_3}{C} = I_2R_2 \quad \varepsilon = I_1R_1 + \frac{q_3}{C} \Rightarrow \frac{\varepsilon}{R_1} = \frac{I_1R_1}{R_1} + \frac{q_3}{R_1C} \qquad I_1 = \frac{\varepsilon}{R_1} - \frac{q_3}{R_1C}$$

Además: $I_1 = I_2 + I_3$ … (1)

$$\frac{\varepsilon}{R_1} - \frac{q_3}{R_1C} = I_2 + I_3$$

$$\frac{\varepsilon}{R_1} - \frac{q_3}{R_1C} = \frac{q_3}{R_2C} + \frac{dq_3}{dt}$$

$$\frac{dq_3}{dt} = \frac{\varepsilon}{R_1} - \frac{q_3}{C}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

Sea: $k = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{R_1+R_2}{R_1R_2}$

$$\frac{dq_3}{dt} = \frac{1}{R_1C}(C\varepsilon - kR_1q_3)$$

$$\frac{dq_3}{C\varepsilon - kR_1q_3} = \frac{dt}{R_1C}$$

Entonces:

$$-\frac{kt}{C} = \left[Ln\frac{(C\varepsilon - kR_1q_3)}{(C\varepsilon)}\right] \qquad e^{-\frac{kt}{C}} = \frac{(C\varepsilon - kR_1q_3)}{(C\varepsilon)}$$

$$q_3 = \frac{C\varepsilon}{kR_1}\left(1 - e^{-\frac{kt}{C}}\right) \Rightarrow \boxed{I_3 = \frac{\varepsilon}{R_1}e^{-\frac{kt}{C}}}$$

4a) Corriente (I_1) por la batería, al instante t=0

De (1): $I_1 = \frac{q_3}{R_2C} + I_3$ … (2)

Cuando t = 0 s, $q_3 = 0 \wedge I_3 = \frac{\varepsilon}{R_1}$

Entonces: $\boxed{I_1(0) = I_3(0) = \frac{\varepsilon}{R_1} = 5\ mA}$

4b) Corriente (I_1) por la batería, al instante t = ∞

De (2): $\boxed{I_1 = \frac{\varepsilon}{R_1 + R_2} = 3{,}3\ \mu A}$

4c) Voltaje final (V) en el condensador

$\boxed{q_3 = \frac{C\varepsilon}{kR_1}}$ $\boxed{V = \frac{q_3}{C} = 9{,}99\ V}$

4d) Gráfica q_3 vs t

$$q_3 = \frac{C\varepsilon}{kR_1}\left(1 - e^{-\frac{kt}{C}}\right) = 3{,}33\mu C\left(1 - e^{-0{,}30t}\right)$$

Donde t está en ms.